

Real-Time Scenario – Using Snowpipe for Automated Data Ingestion

Imagine you are working as a **Data Engineer in an E-commerce company**.

The company website generates **transaction data continuously** whenever customers place orders.

This data must be available in the **Snowflake Data Warehouse in near real-time** so that the analytics team can build dashboards for:

- Live sales monitoring
- Fraud detection
- Customer purchase analysis
- Business performance tracking

Instead of running manual COPY commands repeatedly, the company implements **Snowpipe for automated ingestion**.

End-to-End Architecture

Customer Orders from Website



Application Server Generates Transaction Files



Files Uploaded to **Amazon S3 Bucket**



S3 Event Notification Triggered



Snowpipe Activated



COPY Command Executes Automatically



Data Loaded into **Snowflake Table**



BI Tools (Power BI / Tableau) Read the Data

This pipeline ensures **continuous data loading without manual intervention**.

Step 1 – Create Target Table in Snowflake

First, we create a table where streaming transaction data will be stored.

Example table:

TRANSACTION_ID
CUSTOMER_ID
PRODUCT_ID
ORDER_DATE
QUANTITY
AMOUNT

This table will store all incoming transactions.

Step 2 – Create External Stage

The company stores incoming files in **Amazon S3**.

So we create an external stage in Snowflake pointing to the S3 bucket.

Example concept:

S3 Bucket Location

s3://ecommerce-transactions/orders/

Snowflake stage references this location so it can read files from S3.

Step 3 – Test the COPY Command

Before using Snowpipe, we test the COPY command manually.

The COPY command loads sample files into the Snowflake table.

Purpose of testing:

- Validate file format
- Check column mapping
- Verify data loads correctly

Once successful, this COPY command will be used inside the Snowpipe definition.

Step 4 – Create Snowpipe

Now we create the **Snowpipe object**.

Snowpipe contains the COPY command and defines the automated ingestion logic.

Snowpipe monitors the stage and automatically loads new files into the table.

Important point:

Snowpipe is essentially a **wrapper around the COPY command**.

Step 5 – Configure Cloud Notification

To trigger Snowpipe automatically, we configure **AWS S3 Event Notification**.

Whenever a new file arrives in the S3 bucket:

1. S3 generates an event notification
2. Notification is sent to **SQS Queue**
3. Snowpipe receives the event
4. Snowpipe loads the file into Snowflake

This enables **fully automated near real-time ingestion**.

Step 6 – Upload New Files to S3

Whenever the application generates new files:

Example files:

orders_2026_03_10_01.csv

orders_2026_03_10_02.csv

orders_2026_03_10_03.csv

These files are uploaded automatically to the S3 bucket.

Within a few seconds Snowpipe loads them into the Snowflake table.

Step 7 – Validate Data Load

Data Engineers validate the ingestion.

Example verification queries:

Check row count increase

Check latest transaction records

Confirm ingestion pipeline is working correctly.

Real-Time Benefits of Snowpipe

- Automatic data ingestion
- Near real-time analytics
- No warehouse management
- Serverless scaling
- Reduced operational effort

Snowpipe is widely used for **streaming pipelines and event-driven architectures**.

Real-Time Tools Used in This Project

Application System → Generates files

Amazon S3 → Stores incoming files

AWS Event Notification → Detects new files

Snowpipe → Loads data automatically

Snowflake Table → Stores structured data

BI Tools → Dashboard and analytics

Key Learning from This Project

Snowpipe enables **modern real-time data pipelines** where data automatically flows from applications to analytics platforms without manual intervention.

This architecture is commonly used in:

- E-commerce platforms
- Banking systems
- IoT applications
- Streaming analytics platforms

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